

Dynamic Games of Incomplete (Imperfect) Information

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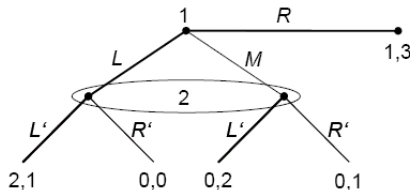
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Dynamic Games of Incomplete (Imperfect) Information

- Players move sequentially.
- Some players do not observe earlier moves of other players in the game.
 - the earlier moves may be by other strategic players;
 - or by nature (nonstrategic) selecting a type for a particular player;
- We will see that SPNE needs to be strengthened.
- With limited information regarding past history, players base their strategies on their beliefs about earlier moves by other players rather than their observation.
- Thus, beliefs will play an important part of the equilibrium characterization.
- We will require that in equilibrium beliefs are consistent with the equilibrium strategies.

Why is SPNE not enough?



- How many subgames are there?
- What are the pure-strategy SPNE of the game?
- Why we may argue that some of them are not reasonable?

Finding Reasonable Refinement: Belief System

Belief System (R1)

A belief system μ in an extensive form game Γ is a specification of a probability $\mu(x|h(x)) \in [0, 1]$ for each decision node $x \in h(x)$ and every h such that

$$\sum_{x \in h(x)} \mu(x|h(x)) = 1$$

- In our example, let $x = L$ and $x' = M$ and $h = \{L, M\}$. Then, we can assign $\mu(L|h) = p$ and $\mu(M|h) = 1 - p$.

Finding Reasonable Refinement: Sequential Rationality (SR)

Sequential Rationality (R2)

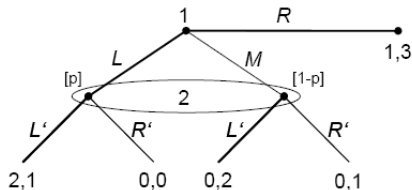
Let $i(h)$ denote the player who moves at an information set h . Then, a strategy profile σ is sequentially rational at information set h if given belief μ ,

$$E[u_{i(h)}|h, \mu, \sigma_{i(h)}, \sigma_{-i(h)}] \geq E[u_{i(h)}|h, \mu, \sigma'_{i(h)}, \sigma_{-i(h)}]$$

for all $\sigma'_{i(h)} \in \Sigma_{i(h)}$. If σ satisfies this for every information set h in Γ , then we say that σ is sequentially rational given μ .

- Therefore, σ is sequentially rational if no player finds it worthwhile to revise her strategy at any point in the game.

Example



- Is it ever sequentially rational for player 2 to player R' ?

Finding Reasonable Refinement: Setting Beliefs Right

Path of the Play

Given an extensive form game Γ and a strategy profile σ , the information set h is *on the path of the play* if it is reached with positive probability.

$$\sum_{x \in h(x)} Pr(x|\sigma) = Pr(h|\sigma) > 0$$

Equilibrium beliefs (R3)

At information sets on the equilibrium path, beliefs are determined using Bayes' rule:

$$Pr(x|h, \sigma) = \frac{Pr(x|\sigma)}{Pr(h|\sigma)}$$

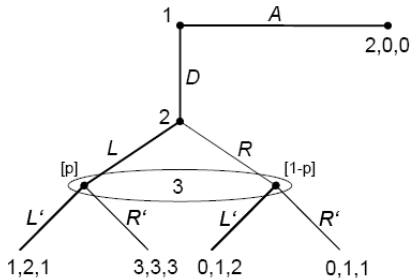
Weak Perfect Bayesian Equilibrium (WPBE)

Definition

Given an extensive-form game Γ , the strategy profile σ and the belief system μ constitutes a WPBE if R1-R3 are satisfied.

- In our example, $\{L, L', p = 1\}$ constitutes a WPBE.
- It is also the unique WPBE.

Example



- What is the SPNE?
- What is the corresponding WPBE?

Further Refinement of Equilibrium Beliefs

Beliefs off the Path of the Play (R4)

Beliefs off the path of the play are derived using Bayes' rule whenever possible.

Definition: Perfect Bayesian Equilibrium

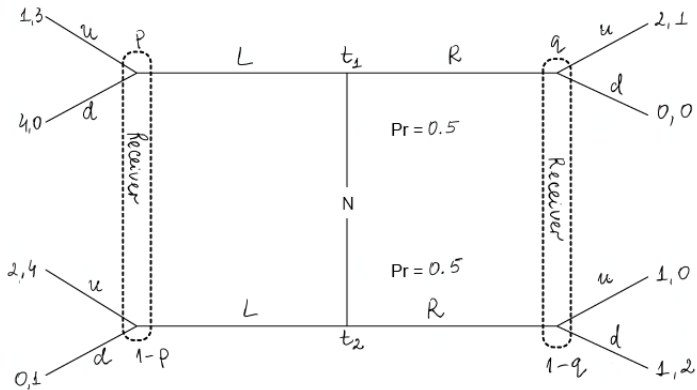
Given an extensive-form game Γ , the strategy profile σ and the belief system μ constitutes a PBE if R1-R4 are satisfied.

- WPBE does not have to be SPNE, but every PBE is SPNE. In fact, PBE is a subset of both WPBE and SPNE.

Signaling game

- 2 players: sender (S) and receiver (R);
- Sender is privately informed about his type;
- Timing:
 - Nature draws a type t_i for the sender from a set of feasible types: $\{t_1, \dots, t_I\}$ according to some distribution $p(\cdot)$ with $p(t_i) > 0$;
 - S observes t_i and chooses from a set of available messages $M = \{m_1, \dots, m_J\}$;
 - R observes m_j (but not t_j) and chooses from a set of available actions $A = \{a_1, \dots, a_K\}$;
 - Payoffs $U_s(t_i, m_j, a_k)$ and $U_R(t_i, m_j, a_k)$ are realized.
- In equilibrium, the message may carry some information about the sender's type.

Example



Pooling and Separating Equilibrium

- How many strategies does the sender have?
- How many strategies does the receiver have?
- A PBE consists of $\{m^*(t_i), a^*(m^*|\mu), \mu\}$.
- A PBE of the signaling game is *pooling* if $m^*(t_1) = m^*(t_2)$. Then $Pr(t_1|m^*) = Pr(t_2|m^*) = \frac{1}{2}$.
- A PBE of the signaling game is *separating* if $m^*(t_1) \neq m^*(t_2)$. Then $Pr(t_i|m^*(t_i)) = 1$;
- Find all pooling and separating PBE.
- An additional option- *semi-pooling*: both types are mixing over the two possible messages, but with different probabilities.